

最优化理论与方法

第2讲 无约束优化问题和约束优化问题

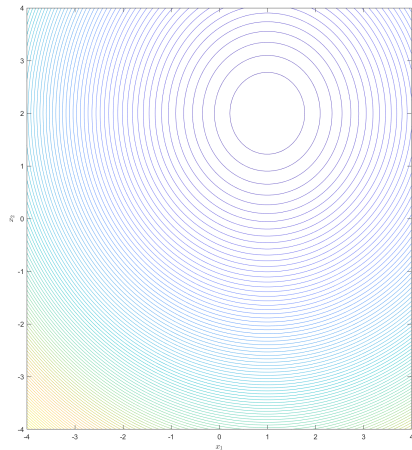
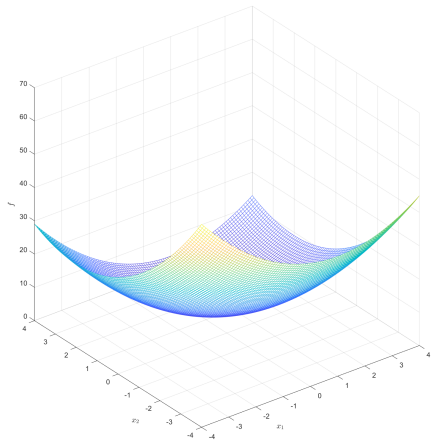
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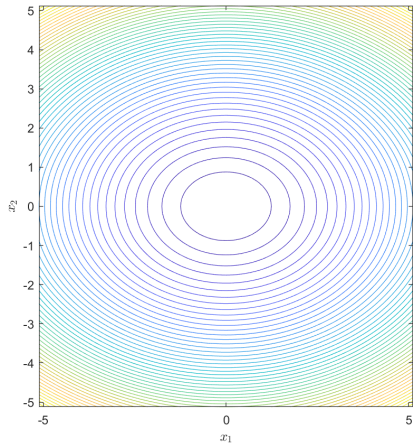
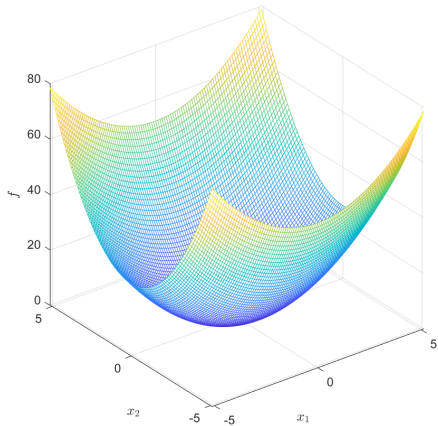
1. 无约束优化问题： 示例

$$f(\mathbf{x}) = (x_1 - 1)^2 + (x_2 - 2)^2$$



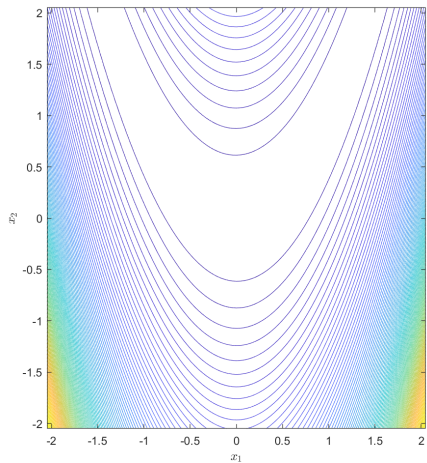
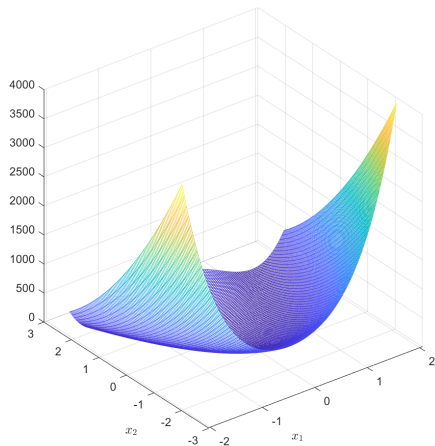
1. 无约束优化问题： 示例

Ellipsoid: $f(\mathbf{x}) = x_1^2 + 2x_2^2$



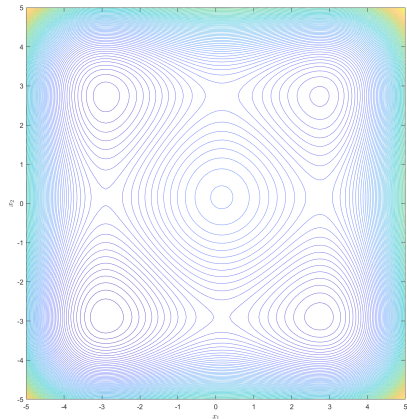
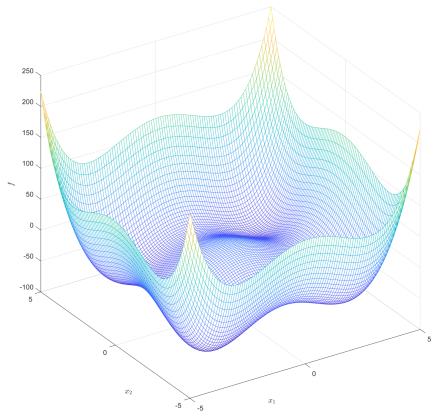
1. 无约束优化问题： 示例

Rosenbrock: $f(\mathbf{x}) = 100(x_2 - x_1^2)^2 + (x_1 - 1)^2$



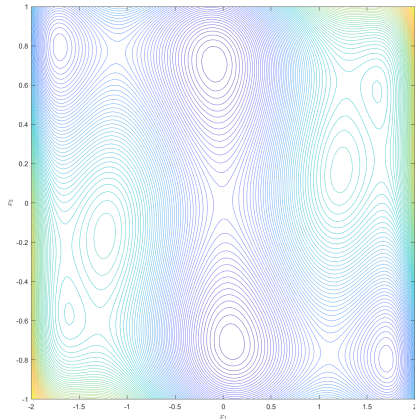
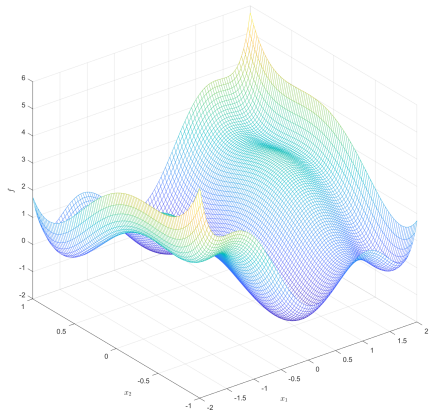
1. 无约束优化问题： 示例

StyblinskiTang: $f(\mathbf{x}) = 0.5 (x_1^4 - 16x_1^2 + 5x_1) + 0.5 (x_2^4 - 16x_2^2 + 5x_2)$



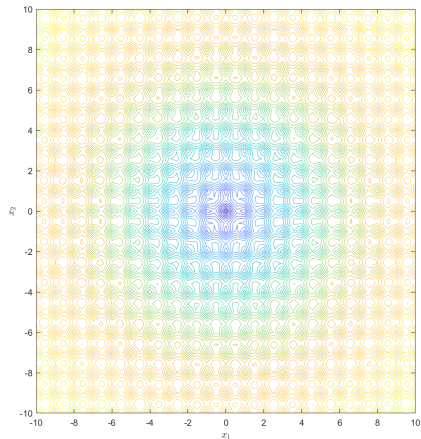
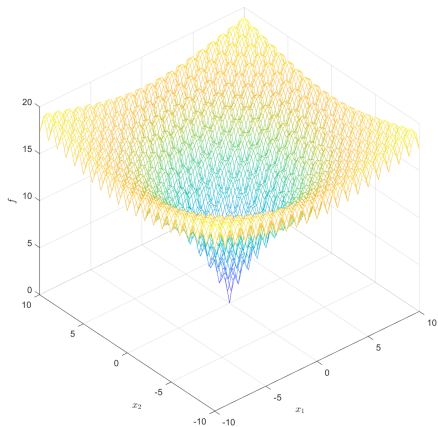
1. 无约束优化问题： 示例

$$\text{SixHump: } f(\mathbf{x}) = \left(4 - 2.1x_1^2 + \frac{x_1^4}{3}\right)x_1^2 + x_1x_2 + (-4 + 4x_2^2)x_2^2$$



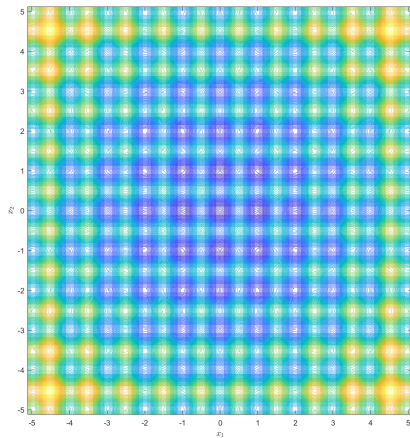
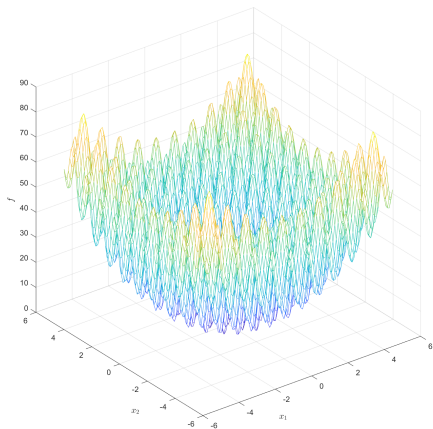
1. 无约束优化问题： 示例

$$\text{Ackley: } f(\mathbf{x}) = -20 \exp \left(-0.2 \sqrt{\frac{x_1^2 + x_2^2}{2}} \right) - \exp \left(\frac{\cos(2\pi x_1) + \cos(2\pi x_2)}{2} \right) + 20 + \exp(1)$$



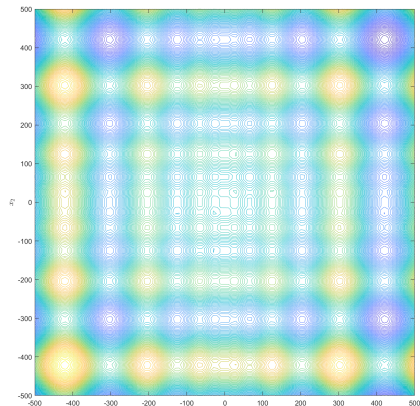
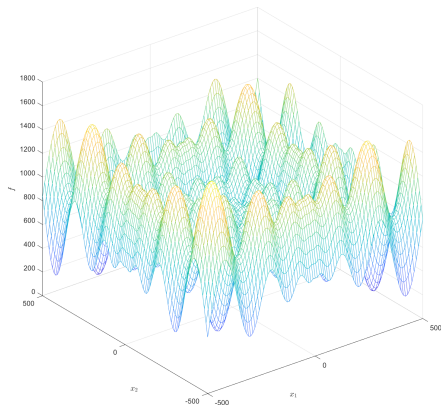
1. 无约束优化问题： 示例

$$\text{Rastrigin: } f(\mathbf{x}) = 20 + (x_1^2 - 10 \cos(2\pi x_1)) + (x_2^2 - 10 \cos(2\pi x_2))$$



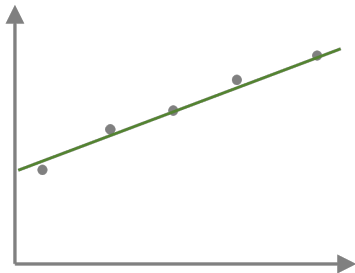
1. 无约束优化问题： 示例

Schwefel: $f(\mathbf{x}) = 837.9658 - x_1 \sin \sqrt{|x_1|} - x_2 \sin \sqrt{|x_2|}$



1. 无约束优化问题：线性回归模型损失函数

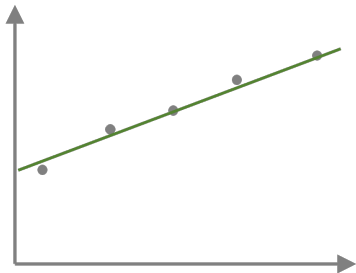
x	y
1	4.5
2	7.1
3	9.9
4	11.9
5	12.3



模型： $f = w_1x + w_2$

1. 无约束优化问题：线性回归模型损失函数

x	y	f
1	4.5	$w_1 + w_2$
2	7.1	$2w_1 + w_2$
3	9.9	$3w_1 + w_2$
4	11.9	$4w_1 + w_2$
5	12.3	$5w_1 + w_2$

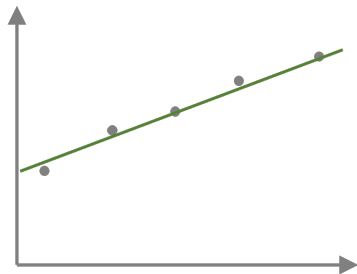


损失函数：

$$\begin{aligned} Loss(w_1, w_2) = & (w_1 + w_2 - 4.5)^2 + (2w_1 + w_2 - 7.1)^2 + (3w_1 + w_2 - 9.9)^2 \\ & + (4w_1 + w_2 - 11.9)^2 + (5w_1 + w_2 - 12.3)^2 \end{aligned}$$

1. 无约束优化问题：线性回归模型损失函数

x	y	f
1	4.5	$w_1 + w_2$
2	7.1	$2w_1 + w_2$
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4	11.9	$4w_1 + w_2$
5	12.3	$5w_1 + w_2$

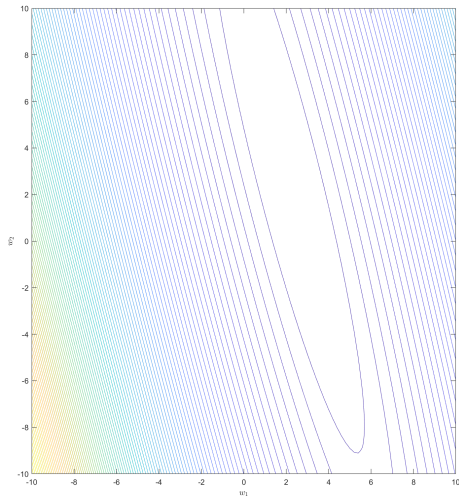
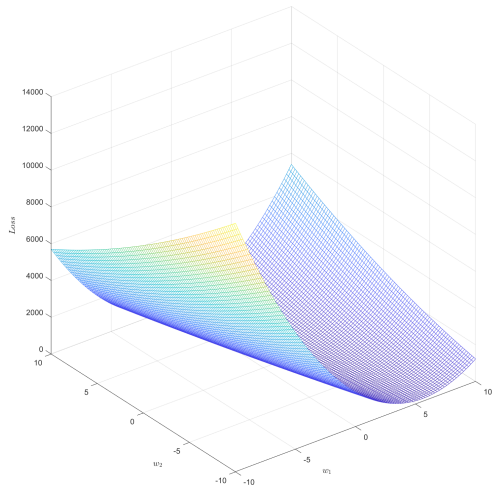


损失函数：

$$Loss(w_1, w_2) = 55w_1^2 + 5w_2^2 + 30w_1w_2 - 315w_1 - 91.4w_2 + 461.57$$

1. 无约束优化问题：线性回归模型损失函数

$$Loss(w_1, w_2) = 55w_1^2 + 5w_2^2 + 30w_1w_2 - 315w_1 - 91.4w_2 + 461.57$$



1. 无约束优化问题：线性回归模型损失函数

(1) 假设模型： $f = w_1x + w_2$

(2) 构建损失： $Loss(w_1, w_2) = 55w_1^2 + 5w_2^2 + 30w_1w_2 - 315w_1 - 91.4w_2 + 461.57$

(3) 求解损失： $\mathbf{w}^* = \operatorname{argmin} Loss(w_1, w_2) = [2.04, 3.04]^T$

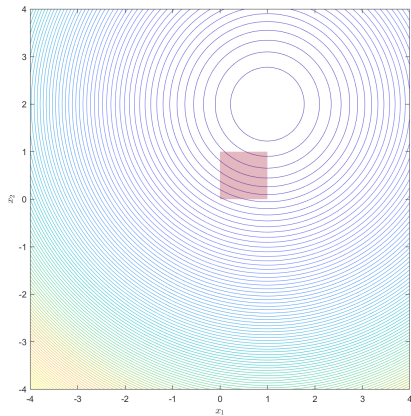
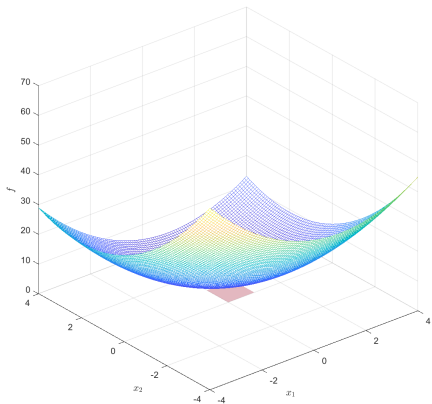
(4) 得到模型： $f = 2.04x + 3.04$

(5) 进行预测： $f_{\text{new}} = 2.04x_{\text{new}} + 3.04$

2. 约束优化问题：边界约束优化问题

$$\text{minimize: } f(\mathbf{x}) = (x_1 - 1)^2 + (x_2 - 2)^2$$

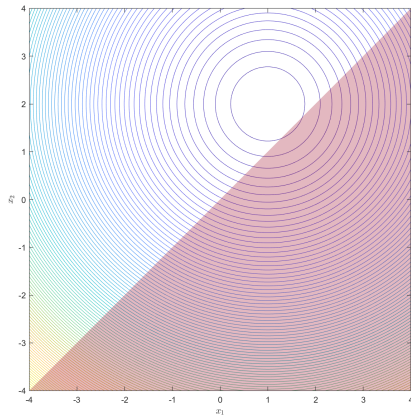
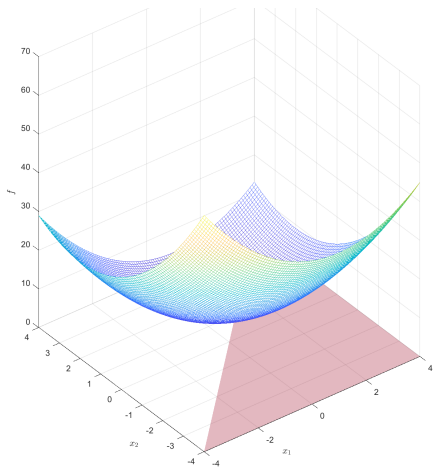
$$\text{subject to: } 0 \leq x_1 \leq 1, \quad 0 \leq x_2 \leq 1$$



2. 约束优化问题： 线性约束优化问题

$$\text{minimize: } f(\mathbf{x}) = (x_1 - 1)^2 + (x_2 - 2)^2$$

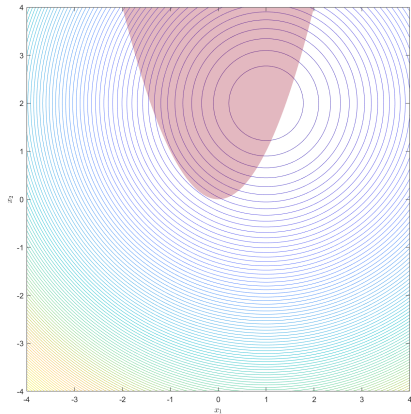
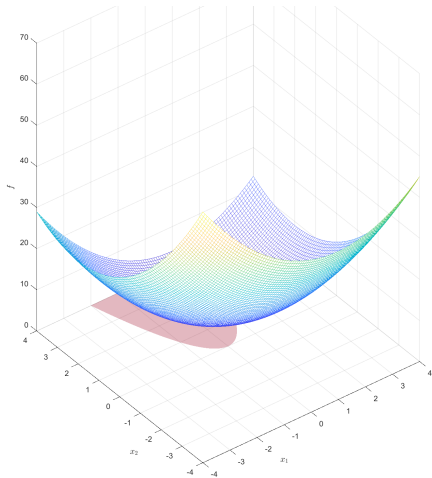
$$\text{subject to: } x_1 \leq x_2$$



2. 约束优化问题：非线性约束优化问题

$$\text{minimize: } f(\mathbf{x}) = (x_1 - 1)^2 + (x_2 - 2)^2$$

$$\text{subject to: } x_1^2 \leq x_2$$

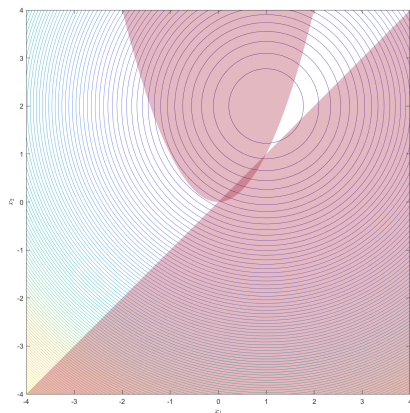
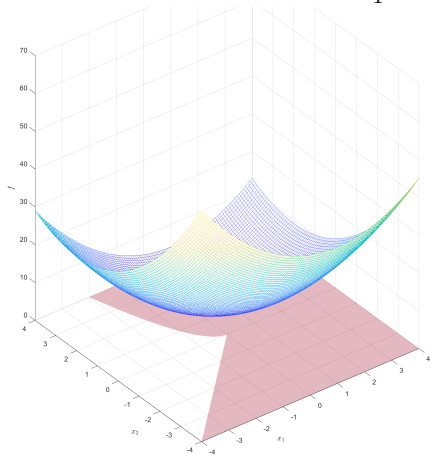


2. 约束优化问题： 复杂约束优化问题

$$\text{minimize: } f(\mathbf{x}) = (x_1 - 1)^2 + (x_2 - 2)^2$$

$$\text{subject to: } x_1 \geq x_2$$

$$x_1^2 \geq x_2$$



2. 约束优化问题：背包优化问题（Knapsack Problem）

假设背包最大承重为 $w_{\max}$ ，如何选取物品使得价值最大？

物品	数量	价值	重量
1	1	v_1	w_1
2	1	v_2	w_2
3	1	v_3	w_3
$\vdots$	1	$\vdots$	$\vdots$
d	1	v_d	w_d

2. 约束优化问题：背包优化问题（Knapsack Problem）

假设背包最大承重为 $w_{\max}$ ，如何选取物品使得价值最大？

物品	数量	价值	重量	变量
1	1	v_1	w_1	x_1
2	1	v_2	w_2	x_2
3	1	v_3	w_3	x_3
$\vdots$	1	$\vdots$	$\vdots$	$\vdots$
d	1	v_d	w_d	x_d

$$\text{maximize: } v(\mathbf{x}) = v_1x_1 + v_2x_2 + \cdots + v_dx_d$$

$$\text{subject to: } w_1x_1 + w_2x_2 + \cdots + w_dx_d \leq w_{\max}$$

$$x_i \in \{0, 1\}, i = 1, 2, \cdots, d$$

2. 约束优化问题：背包优化问题（Knapsack Problem）

假设每个物品的数量是多个，优化问题该如何建立？

物品	数量	价值	重量	变量
1	n_1	v_1	w_1	x_1
2	n_2	v_2	w_2	x_2
3	n_3	v_3	w_3	x_3
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
d	n_d	v_d	w_d	x_d

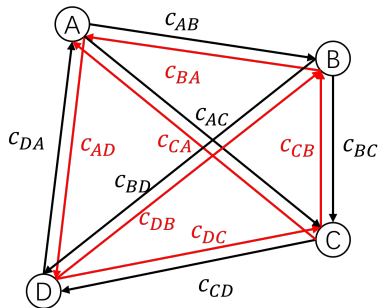
$$\text{maximize: } v(\mathbf{x}) = v_1x_1 + v_2x_2 + \cdots + v_dx_d$$

$$\text{subject to: } w_1x_1 + w_2x_2 + \cdots + w_dx_d \leq w_{\max}$$

$$x_i \in \{0, 1, \cdots, n_d\}, i = 1, 2, \cdots, d$$

2. 约束优化问题：旅行商问题 (Traveling Salesman Problem)

要求经过每个城市且只经过一次，如何规划路线使得代价最小？



$$\mathbf{x} = [x_{AB}, x_{AC}, \dots, x_{DA}, x_{DB}, x_{DC}]^T$$

$$\text{minimize: } c(\mathbf{x}) = c_{AB}x_{AB} + \dots + c_{DC}x_{DC}$$

$$\text{subject to: } x_{BA} + x_{CA} + x_{DA} = 1$$

$$x_{AB} + x_{AC} + x_{AD} = 1$$

$$\vdots$$

$$x_{AD} + x_{BD} + x_{CD} = 1$$

$$x_{DA} + x_{DB} + x_{DC} = 1$$

$$x_{ij} \in \{0, 1\}$$

如果有 d 个城市，总组合数是 $n = (d-1)!$ 。

当 $d = 20$ 时， $n = 19! \approx 1 \times 10^{17}$ 。